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B.Tech. DEGREE EXAMINATION, DECEMBER 2022
Third Semester

18BTC101J - BIOCHEMISTRY

(For the candidates admitted from the academic year 2020-2021 to 2021-2022)

Note:

- (i) **Part - A** should be answered in OMR sheet within first 40 minutes and OMR sheet should be handed over to hall invigilator at the end of 40th minute.
- (ii) **Part - B** should be answered in answer booklet.

Time: 2½ Hours

Max. Marks: 75

PART – B (5 × 10 = 50 Marks)

Answer **ALL** Questions

	Marks	BL	CO	PO
26. a. Write an essay on the different types of polysaccharide. Add a short note on lectins.	10	3	1	1,1 3
(OR)				
b. Explain the structure and organization of proteins, and mention the chemical bonds involved in it.	10	3	1	1,1 3
27. a. Explain the biochemical reactions that occur in TCA cycle and describe how the citric acid cycle is regulated.	10	4	1	1,4 ,13
(OR)				
b. Classify the diabetes mellitus and list out the complications associated with them. Add a note on insulin release in the pancreas.	10	4	1	1,4 ,13
28. a. Write an essay on the important biomolecules that are derived from amino acids.	10	3	3	1,4 ,14
(OR)				
b. Solve the step by step process of how the two nitrogen from amino acids are removed as urea in our body.	10	3	3	1,4 ,14
29. a. List out the steps and identify the repeated steps of fatty acids degradation.	10	4	4	1,4 ,15
(OR)				
b. Describe the pathway through which purine nucleotide synthesis occurs.	10	4	4	1,4 ,15
30. a. Describe the key components of the electron transport chain, and add notes on how they are arranged?	10	4	5	4,5 ,15
(OR)				
b. Describe how the light reactions and the Calvin cycle are coordinated.	10	4	5	4,5 ,15

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